



Targeting in the Presence of Noise

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Abstract—Targeting methods direct a trajectory from an initial state to a chosen final state in a few iterations by making a small perturbation at the first step. Existing methods have two drawbacks: they require that the map is known so that it can be back iterated from the target region and they generally fail in the presence of noise. We address the first problem by collecting the history of iterations which fall in the target region. To counteract the effects of noise, an averaged idealised trajectory is determined together with a linearisation of the map around this trajectory. A small parameter perturbation can then be made at each stage of the targeting process to keep the trajectory close to the idealised one. These methods are illustrated by an example. © 1998 Elsevier Science Ltd. All rights reserved

1. INTRODUCTION

Since the pioneering paper of Ott *et al.* [1] much attention has been paid to the control of chaos via parametric perturbations. Their method involves varying a system parameter of a chaotic system in order to stabilise an unstable periodic orbit contained in the chaotic attractor. This is achieved by using a parameter perturbation to move iterates across onto the stable manifold of the periodic orbit. In many physical systems, chaotic motion may be undesirable and so control methods provide a simple route to regular motion and require only small perturbations to the system. Parametric perturbations can also be applied to systems in order to achieve other goals. These include targeting [2–7] whereby trajectories are directed towards specific targets by the application of small, discerningly chosen perturbations of an available system parameter.

One of the problems associated with the control of a system is that lengthy transients are often observed before control can be fully effected. Generally, the smaller the maximal allowed control signal, the longer the expected transient length [1]. Steps can be taken to reduce these transient lengths. One approach is to increase the size of the basin of attraction of the control method. This can be achieved by increasing the size of $\delta p_{\max}$, the maximum allowed parameter perturbation, although this may prove relatively ineffective since $\delta p_{\max}$ must always be ‘small’. A more recent approach [8] extends the basin of attraction for the control method without increasing $\delta p_{\max}$ by requiring that the n th iterate be placed on the stable manifold rather than the first, for some $n > 1$.

A more effective way of reducing transient lengths is by using targeting. This method provides us with a valuable tool for changing the state of an ergodic process from an initial state to some desired target state in a much shorter time than if no perturbation were applied to the system. Consequently, for example, this gives us the potential to trim efficiently the average length of the chaotic transient typically experienced before a system can be brought under control via the application of some form of control method. The

targeting technique could also provide us with an efficient means of switching from one desired behaviour to another, e.g. switching from one controlled periodic point to another. The effectiveness of targeting is due to the sensitive dependence of chaotic systems to small perturbations. Thus, the problem is to find an appropriate perturbation in order to achieve a desired effect.

Provided that the dynamical equations of the system are known, or a good (global) approximation to them can be obtained, the method of Shinbrot *et al.* [5] can be used to achieve both of the goals discussed above. If the equations are not known, the method is not immediately applicable. However, we derive a modified method of targeting which does not require any knowledge of the underlying equations or any global approximation to them.

Targeting methods usually work well in the absence of noise but they will generally fail if there is noise in the system. In [5], this problem was overcome by calculating a perturbation which, if applied at the first step, would direct the trajectory to the target in a small number of iterations in the system without noise. When there is noise present, then only the first step is taken and then the same problem is solved again from the new position and a new perturbation is derived and applied. This procedure is applied at every iteration, thus requiring a large amount of computation at each step.

An alternative method of targeting is described in [2] using an open loop controller where the controls are derived using optimal control theory. In systems without noise, all the controls can be computed in advance. However, in the presence of noise, again only the first step is taken and the whole problem is solved from the new position. This process is repeated at each iteration and so again a large amount of computation is required at each step. A similar method of targeting is considered in [7] but the effect of noise is not addressed in this case.

In order to avoid having to perform large amounts of computation on line as part of the targeting process, we consider an alternative method of dealing with noise. This method consists of constructing an averaged, idealised trajectory based on one initial parameter perturbation. During the targeting process, small parameter perturbations are made to keep the iterates close to this idealised trajectory. The calculation of these perturbations requires an approximation to the linearised map at each step, but these approximations can be constructed off line, before the targeting process begins. Thus, there is only a small calculation required to determine the appropriate parameter perturbation at each step of the targeting process.

In Section 2, we review the method of Shinbrot *et al.* [5] for targeting when the equations are known. A modification of this method when the equations are not known is described in Section 3. In Section 4, the use of parameter perturbations to counteract the effects of noise is described and these methods are all illustrated in a numerical example in Section 5.

2. TARGETING WHEN THE EQUATIONS ARE KNOWN

We consider systems of the form

$$\mathbf{x}_{n+1} = \mathbf{F}(\mathbf{x}_n, p), \quad (1)$$

where $\mathbf{F}: \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}^2$ and p is a system parameter available for tuning at each iteration of the map about some nominal value p^* . Parameter perturbations are restricted by some maximum value $\delta p_{\max} > 0$ such that $|\delta p_n| \leq \delta p_{\max}$, where $\delta p_n = p_n - p^*$.

We now briefly review the method proposed by Shinbrot *et al.* [5] for targeting in systems where the mapping $\mathbf{F}$ is known. The aim is to proceed from an initial state $\mathbf{x}_i$ to the vicinity of a target state $\mathbf{x}_t$ in a small number of steps. After one iteration of the map (1), beginning

with the initial condition $\mathbf{x}_0 = \mathbf{x}_s$, the change in the state of the system due to a perturbation of δp_0 applied at the first step is given by

$$\delta \mathbf{x}_1 := \mathbf{F}(\mathbf{x}_s, p^* + \delta p_0) - \mathbf{F}(\mathbf{x}_s, p^*) = D_p \mathbf{F}(\mathbf{x}_s, p^*) \delta p_0, \quad (2)$$

ignoring terms of $O(\delta p_0^2)$. Letting δp_0 vary through its permissible range of values $-\delta p_{\max} \leq \delta p_0 \leq \delta p_{\max}$, eqn (2) defines a small line segment through $\mathbf{x}_1 = \mathbf{F}(\mathbf{x}_s, p^*)$. Denote this line segment by $\overline{\delta x}$ and its length by δx .

With each successive iteration of the map (1), the line segment will grow in length approximately geometrically at a rate governed by the positive Lyapunov exponent of the system, denoted by λ_1 . Let n_1 be the number of iterations required for the line segment to grow to a length of order 1, which typically happens when $\delta x \exp(n_1 \lambda_1) \sim 1$, for small δx . Without loss of generality, we may take the size of the relevant ergodic region (i.e. the region in which the natural measure of the attractor is defined) to be of the order of 1 so that n_1 is the number of iterations of (1) such that $\overline{\delta x}$ spans the ergodic region.

Define λ_2 to be the negative Lyapunov exponent of the system $\mathbf{F}$ and let $\bar{\epsilon}_1$ be the region about $\mathbf{x}_1$ that we are targeting. Mapping this region backwards in time, its n_2 th preimage will span the ergodic region when $n_2 \sim (1/|\lambda_2|) \ln(1/\epsilon_1)$, where ϵ_1 (assumed small) is a measure of the linear size of $\bar{\epsilon}_1$, e.g. $\bar{\epsilon}_1$ may be a box of side length ϵ_1 . We now look for an intersection of the forward iterated line segment and the backward iterated region. Taking a point on the middle of this intersection and iterating it backward in time n_1 steps, we find a point on the line segment $\overline{\delta x}$ from which we may determine a value of δp_0 which, if applied to the system, beginning with the initial point $\mathbf{x}_s$, should take us to the vicinity of $\mathbf{x}_1$ in $n_1 + n_2$ steps. In the absence of noise, no further perturbations are necessary.

In practice, we cannot iterate a line or a region, so we iterate discrete approximations to them, joining the points with a straight line. Once an intersection is detected, its accuracy may be refined by repeatedly halving the intersecting forward and backward line segments and determining which of these halves actually contains the intersection.

3. TARGETING WHEN THE EQUATIONS ARE UNKNOWN

The targeting algorithm outlined above has been shown to work well if the dynamical equations are known, or if the system is described reasonably accurately by a one-dimensional map [6, 3]. We now describe a targeting algorithm for directing trajectories where the underlying dynamical equations are unknown and are not easily estimated.

Suppose, as before, we wish to proceed from some initial state $\mathbf{x}_s$ to the vicinity of the target $\mathbf{x}_t$, but this time $\mathbf{F}$ is unknown. We cannot iterate a region about the target backward in time without knowledge of $\mathbf{F}$. Instead we let our system run, unperturbed, collecting iterates which we shall label $\mathbf{x}^i$ in an ϵ_t -neighbourhood of $\mathbf{x}_t$. Such collected points satisfy

$$\|\mathbf{x}^i - \mathbf{x}_t\| < \epsilon_t, \quad (3)$$

for small ϵ . Let there be N_b such points $\mathbf{x}^i$. We then form the $2 \times (n_b + 1)$ matrices

$$\mathbf{y}^i = (\mathbf{x}_n^i \quad \mathbf{x}_{n-1}^i \quad \cdots \quad \mathbf{x}_{n-n_b}^i), \quad i = 1, 2, \dots, N_b,$$

where $\mathbf{x}_{k-1}^i$ is the preimage of $\mathbf{x}_k^i$, for $k = n - n_b + 1, \dots, n$, where $n_b > 1$. Such matrices will be called history matrices. Thus, as well as collecting points satisfying (3), we have also collected their preimages to form their history going back n_b steps. Each $\mathbf{y}^i$ is a $2 \times (n_b + 1)$ matrix whose first column defines a point lying in an ϵ -neighbourhood of $\mathbf{x}_t$ and whose remaining columns define a short 'history' of that point. We then start our system running, many times, beginning with the initial condition $\mathbf{x}_s$, each time with a different perturbation

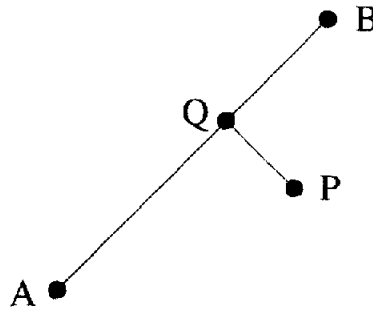


Fig. 1. Finding an approximate intersection.

at the first step, and allow the system to run for at least another n_f steps. We shall indicate how to determine n_b and n_f presently.

Suppose that the applied perturbations are spaced evenly in the interval $[-\delta p_{\max}, \delta p_{\max}]$ (although this is not a necessary requirement) and let there be N_f such perturbations. Then we can form the approximation to the line segment $\bar{\delta x}$ as given by eqn (2), together with its n_f forward images, by simply joining neighbouring points with straight line segments. We then look for 'backward iterated' points lying on, or at least very close to, the forward iterated line segment $\bar{\delta x}$. Labelling the perturbations as $-\delta p_{\max} \leq \delta p_0^1 < \delta p_0^2 < \dots < \delta p_0^{N_f} \leq \delta p_{\max}$, we say that an intersection between the j th forward iterated line segment (corresponding to the forward iterated points with first step perturbations of δp_0^j and δp_0^{j+1}) and a 'backward iterated' trajectory has occurred if $\mathbf{x}_{n-n_b}^j$ approximately lies on this line, for some i .

Let $A = \mathbf{x}_{n_f}(\delta p_0^j)$ be the n_f th point on the forward trajectory corresponding to a first step perturbation of δp_0^j . Let $B = \mathbf{x}_{n_f}(\delta p_0^{j+1})$ be similarly defined, and let P be the point between A and B of the (approximately) intersecting history matrix, $P = \mathbf{x}_{n-n_b}^j$. An intersection is determined as follows (see Fig. 1). Form the vector

$$\mathbf{d}_i = \overrightarrow{AB} = (\mathbf{x}_{n_f}(\delta p_0^j) - \mathbf{x}_{n_f}(\delta p_0^{j+1}))$$

and the perpendicular $\mathbf{d}_f^\perp$ to this vector, passing through the point P . Let Q be the point where $\mathbf{d}_i$ meets $\mathbf{d}_f^\perp$. Then an intersection is deemed to have occurred when

$$\|\overrightarrow{PQ}\| < \epsilon,$$

for some suitably small $\epsilon > 0$. The initial parameter perturbation required to obtain a forward iterate close to the point Q can then be found using linear interpolation, giving

$$\delta p_0 = \delta p_0^j + \frac{\|\overrightarrow{AQ}\|}{\|\overrightarrow{AB}\|} (\delta p_0^{j+1} - \delta p_0^j). \quad (4)$$

Thus, given $\mathbf{x}_s$ and a first step perturbation of δp_0 defined by eqn (4), we can form the trajectory

$$\mathbf{Y}_1 = (\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_{n_f+n_b}),$$

where $\mathbf{x}_1 = \mathbf{F}(\mathbf{x}_s, \delta p_0)$ and $\mathbf{x}_k = \mathbf{F}(\mathbf{x}_{k-1}, p^*)$, for $k = 2, 3, \dots, n_f + n_b$. We shall call this the idealised trajectory from $\mathbf{x}_s$ to the vicinity of $\mathbf{x}_1$. So if we start at $\mathbf{x}_s$, make a small perturbation of δp_0 to the system at the first step, we will then follow the idealised trajectory to the vicinity of $\mathbf{x}_1$ in $n_f + n_b$ steps.

We remark that the targets we aim for using this algorithm must necessarily lie on the attractor and consequently, if n_f is not to be too large, $\mathbf{x}_s$ should also lie on or near the

attractor. The algorithm is thus particularly well suited to switching between different states lying on the attractor.

4. THE EFFECT OF NOISE

The algorithm described in the previous section is not robust, since even very small amplitude noise can cause an observed trajectory to wander far from the idealised trajectory during the targeting process (see the numerical example of Section 5). This is clearly due to the extreme sensitivity of chaotic systems to small perturbations. In this section, we propose a method which helps combat the detrimental effect of small amplitude noise on our algorithm. The method consists of finding an averaged idealised trajectory from $\mathbf{x}_s$ to the vicinity of $\mathbf{x}_t$. That is, we determine a trajectory that takes us from $\mathbf{x}_s$ to $\mathbf{x}_t$ which is an average of several forward and several 'backward iterated' trajectories that approximately intersect at some point on the attractor. We then estimate the linearised maps at each point on the averaged idealised trajectory and use these estimates to apply a perturbation at each step such that the observed trajectory approximately follows the averaged idealised trajectory from $\mathbf{x}_s$ to the vicinity of $\mathbf{x}_t$. Note that all the calculations required for this method can be done in advance so that there is very little computation required during the actual targeting process.

The first step is the same as before. We determine an intersection of a forward iterated line segment and a 'backward iterated' point and estimate δp_0 . Denote the point at which the intersection occurs by $\mathbf{x}_{\text{int}}$. We then take an ϵ -neighbourhood about $\mathbf{x}_{\text{int}}$ and determine m_f forward trajectories emanating from $\mathbf{x}_s$ with some perturbation at the first step whose n_f th point lies within this neighbourhood. Numerically, one determines m_f points (and their histories) on trajectories emanating from $\mathbf{x}_s$ such that

$$\|\mathbf{x}_{n_f} - \mathbf{x}_{\text{int}}\| < \epsilon,$$

for some small $\epsilon > 0$. Denote these m_f trajectories by

$$\mathbf{X}_f^k = (\mathbf{x}_{1f}^k \quad \mathbf{x}_{2f}^k \quad \cdots \quad \mathbf{x}_{n_f}^k), \quad k = 1, 2, \dots, m_f,$$

where $(\mathbf{x}_j^k)^T = (x_j^k, y_j^k)$ are the coordinates of the j th point on the k th trajectory. We average these trajectories to form the approximate forward part of the idealised trajectory so that

$$\bar{\mathbf{x}}_{jf} = \begin{pmatrix} \bar{x}_j \\ \bar{y}_j \end{pmatrix} = \begin{pmatrix} (1/m_f) \sum_{k=1}^{m_f} x_j^k \\ (1/m_f) \sum_{k=1}^{m_f} y_j^k \end{pmatrix}, \quad j = 1, 2, \dots, n_f. \quad (5)$$

In doing so we hope to average out the noise. Similarly averaging the first step perturbations gives

$$\overline{\delta p_0} = \frac{1}{m_f} \sum_{k=1}^{m_f} \delta p_0^k.$$

As before, we collect iterates in the neighbourhood of the target point $\mathbf{x}_t$ together with their histories. We then average the m_b backward trajectories whose n_b th point lies in an ϵ -neighbourhood of $\mathbf{x}_{\text{int}}$ so that

$$\bar{\mathbf{x}}_{lb} = \begin{pmatrix} \bar{x}_l \\ \bar{y}_l \end{pmatrix} = \begin{pmatrix} (1/m_b) \sum_{k=1}^{m_b} x_l^k \\ (1/m_b) \sum_{k=1}^{m_b} y_l^k \end{pmatrix}, \quad l = 1, 2, \dots, n_b. \quad (6)$$

We can then form the approximate idealised trajectory from $\mathbf{x}_s$ to the vicinity of $\mathbf{x}_t$ by concatenating the $\bar{\mathbf{x}}_{jf}$, for $j = 1, 2, \dots, n_f$ given by eqn (5) with the $\bar{\mathbf{x}}_{lb}$, for $l = n_b, n_b - 1, \dots, 1$ of eqn (6), giving the averaged idealised trajectory $\bar{\mathbf{Y}}_1$ as

$$\bar{\mathbf{Y}}_1 = (\bar{\mathbf{x}}_{1f} \quad \bar{\mathbf{x}}_{2f} \quad \cdots \quad \bar{\mathbf{x}}_{n_f} \quad \bar{\mathbf{x}}_{(n_b-1)b} \quad \cdots \quad \bar{\mathbf{x}}_{1b}).$$

We now suppose that during the targeting process, the n th iterate $\mathbf{x}_n$ is close to the n th point $\bar{\mathbf{Y}}_{1n}$ of the averaged idealised trajectory $\bar{\mathbf{Y}}_1$. We then want to compute a new value p_n of the parameter which ensures that $\mathbf{x}_{n+1}$ and $\bar{\mathbf{Y}}_{1(n+1)}$ are also close. To do this, we need an approximation to the linearised map around the idealised trajectory. Thus, expanding about the point $(\bar{\mathbf{Y}}_{1n}, p^*)$, we get

$$\begin{aligned}\mathbf{x}_{n+1} &= \mathbf{F}(\bar{\mathbf{Y}}_{1n} + (\mathbf{x}_n - \bar{\mathbf{Y}}_{1n}), p^* + (p_n - p^*)) \\ &= \mathbf{F}(\bar{\mathbf{Y}}_{1n}, p^*) + D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{Y}}_{1n}, p^*)(\mathbf{x}_n - \bar{\mathbf{Y}}_{1n}) + D_p\mathbf{F}(\bar{\mathbf{Y}}_{1n}, p^*)(p_n - p^*) + \dots \\ &\approx \bar{\mathbf{Y}}_{1(n+1)} + A_n\delta\mathbf{x}_n + \mathbf{w}_n\delta p_n,\end{aligned}$$

where $A_n = D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{Y}}_{1n}, p^*)$, $\mathbf{w}_n = D_p\mathbf{F}(\bar{\mathbf{Y}}_{1n}, p^*)$ and $\delta\mathbf{x}_n = \mathbf{x}_n - \bar{\mathbf{Y}}_{1n}$, which we have assumed to be small. We have also assumed that $\bar{\mathbf{Y}}_{1(n+1)} = \mathbf{F}(\bar{\mathbf{Y}}_{1n}, p^*)$, which does not hold precisely but is a good approximation provided that the noise signal is small. Thus the linearised map about the point $\bar{\mathbf{Y}}_{1n}$ on the idealised trajectory has the form

$$\mathbf{x}_{n+1} = A_n\mathbf{x}_n + \mathbf{b}_n + \mathbf{w}_n\delta p_n,$$

where $\mathbf{b}_n = \bar{\mathbf{Y}}_{1(n+1)} - A_n\bar{\mathbf{Y}}_{1n}$. The A_n and $\mathbf{b}_n$ can be experimentally obtained by setting $\delta p_n = 0$ and letting the system run, collecting data about each point of the approximate idealised trajectory and performing a regression to estimate the local linearised dynamics about each point on the trajectory [9]. Clearly, all the trajectories which were used to find the averaged idealised trajectory can be used in this context also.

The $\mathbf{w}_n$ can be estimated by allowing the system to run, unperturbed, waiting until the state of the system falls within an ϵ -neighbourhood of the n th point on the approximate idealised trajectory, and then perturbing the parameter by some small prescribed value δp_w and finding where the next point $\bar{\mathbf{Y}}_{1(n+1)}(\delta p_w)$ lies on the attractor. Doing this many times for each point n of the trajectory and averaging, we get $\bar{\mathbf{Y}}_{1(n+1)}(\delta p_w)$, and $\mathbf{w}_n$ is then given by

$$\mathbf{w}_n \approx \frac{\bar{\mathbf{Y}}_{1(n+1)}(\delta p_w) - \bar{\mathbf{Y}}_{1(n+1)}}{\delta p_w}.$$

Alternatively, one could use a central difference approximation here. Calculating the $n_f + n_b$ derivative vectors $\mathbf{w}_n$ requires a considerable amount of computation but again we emphasize that these calculations can be performed off line.

The modified targeting algorithm to counteract the effects of noise then consists of applying a perturbation of $\bar{\delta p}_0$ at the first step and applying perturbations δp_n at subsequent steps such that

$$\|\mathbf{x}_{n+1} - \bar{\mathbf{Y}}_{1(n+1)}\|_2^2$$

is a minimum. We thus choose δp_n to minimise

$$\|A_n\mathbf{x}_n + \mathbf{b}_n + \mathbf{w}_n\delta p_n - \bar{\mathbf{Y}}_{1(n+1)}\|_2^2,$$

giving

$$\delta p_n = -\frac{\mathbf{w}_n^T(A_n\mathbf{x}_n + \mathbf{b}_n - \bar{\mathbf{Y}}_{1(n+1)})}{\mathbf{w}_n^T\mathbf{w}_n} = -\frac{\mathbf{w}_n^TA_n(\mathbf{x}_n - \bar{\mathbf{Y}}_{1n})}{\mathbf{w}_n^T\mathbf{w}_n}. \quad (7)$$

Clearly δp_n will be small provided that $\mathbf{x}_n - \bar{\mathbf{Y}}_{1n}$ is small. This is similar to the shortest distance method for stabilising a fixed point using parameter perturbations [10, 9, 11]. One possibility for the determination of n_f and n_b is to use the same criterion as used to determine n_1 and n_2 in the targeting method of Shinbrot *et al.* (see Section 2 and [5]). However, numerical simulations suggest that these estimates can be too large since it is

often the case that we do not need to iterate the line segment and 'back iterate' the target region until they are of order one to find an intersection. The intersection often happens before the iterated regions reach order one. In the numerical example below, we have chosen n_t and n_b in an ad hoc fashion, trying many combinations for which we chose a combination giving the smallest value of $n_t + n_b$, and hence the shortest targeting time. However, n_t and n_b are always chosen such that $n_t \leq n_1$ and $n_b \leq n_2$.

5. NUMERICAL EXAMPLE

As an example, we consider the Hénon map

$$\begin{aligned} x_{n+1} &= 1 - Ax_n^2 + y_n, \\ y_{n+1} &= Bx_n. \end{aligned} \quad (8)$$

We take $A = 1.4$ and $B = 0.3$ as the nominal parameter values at which (8) possesses a chaotic attractor and we choose A as our control parameter. Suppose we wish to progress from the initial state $\mathbf{x}_s = (0.975897, -0.142725)$, very close to a period-2 point, to the vicinity of $\mathbf{x}_t = (0.631355, 0.189406)$, a fixed point. We say that an iterate $\mathbf{x}_n$ is in the vicinity of $\mathbf{x}_t$ when $\|\mathbf{x}_n - \mathbf{x}_t\| < 5 \times 10^{-3}$. Then, without applying the targeting algorithm, 4440 iterations of the Hénon map are required to reach the vicinity of $\mathbf{x}_t$. Letting $\delta p_{\max} = 0.05$ and applying our algorithm, we obtain a predicted value of the first step perturbation of $\delta p_0 = -0.0473$ to target $\mathbf{x}_t$ in only eleven steps. In practice, the distance from $\mathbf{x}_t$ after eleven iterations with this perturbation is 9.4×10^{-3} which is not inside our target region but is close to it.

We now add some noise to the system such that the equations become

$$\begin{aligned} x_{n+1} &= 1 - (1.4 + \delta p_n)x_n^2 + y_n + \xi_n, \\ y_{n+1} &= 0.3x_n + \eta_n, \end{aligned}$$

where the ξ_n and η_n are noise terms, the noise being uniformly distributed in the interval $[-4 \times 10^{-3}, 4 \times 10^{-3}]$. The resulting chaotic attractor is shown in Fig. 2. Applying the new

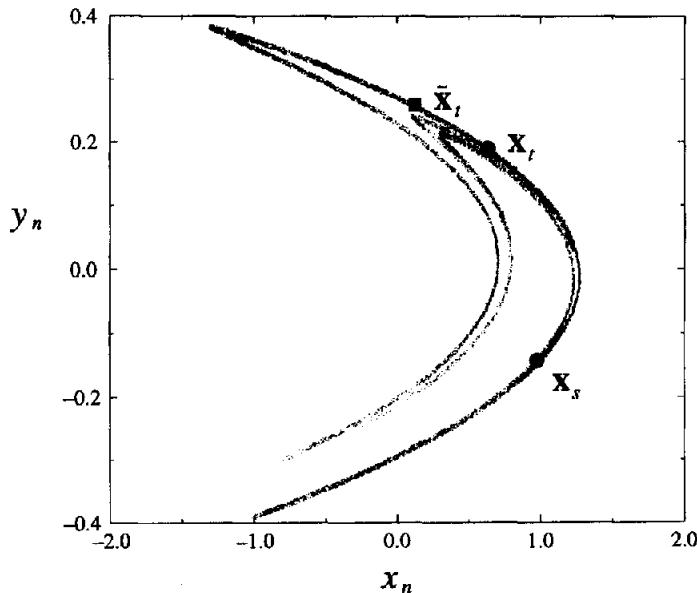


Fig. 2. Targeting in the Hénon map with added noise.

algorithm to find the averaged idealised trajectory, we find that it should be possible to reach $\mathbf{x}_t$ in eleven steps with a first step perturbation of $\delta p_0 = -0.04716$, which is close to the value of the initial perturbation obtained previously without noise. If we apply this perturbation at the first step, and no further perturbations, the trajectory ends up at the point denoted by $\tilde{\mathbf{x}}_t$ shown in Fig. 2 after eleven steps. Estimating the linearised maps at each point on the idealised trajectory and applying a perturbation as prescribed by eqn (7) at each step after the first, we end up a distance of 5.01×10^{-3} from the target after eleven iterations of the map, which is very close to the vicinity of $\mathbf{x}_t$. In producing these results, the Hénon map has only been used to create data. It has not been used explicitly in the targeting process.

To demonstrate the speed at which switching between different controlled states can be effected, we attempt to repeatedly switch between a fixed point and period-2 points.

Using $\delta p_{\max} = 0.02$, we apply the control alternately in the vicinities of the fixed point and period-2 points, applying this control every 3000 iterations of the map. We then repeat the process, but this time using the targeting algorithm between successive switches. The length of the chaotic transients exhibited before periodic motion is achieved is significantly reduced (see Fig. 3). Similar results can be achieved in the presence of small amplitude noise. Note

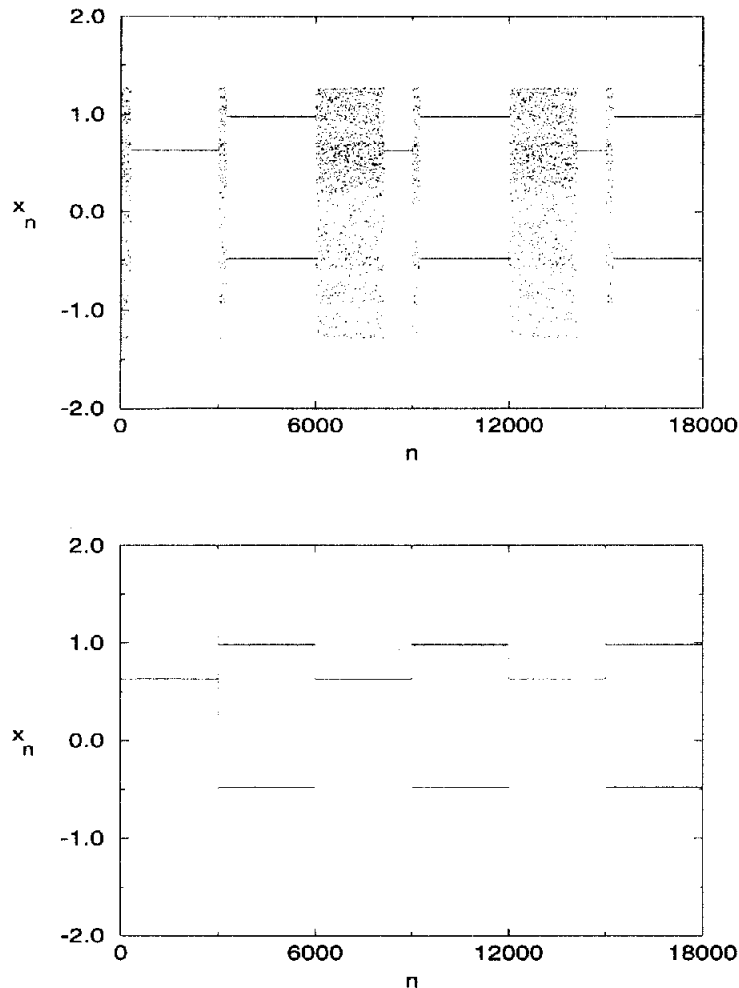


Fig. 3. Switching between controlled states without (top) and with (bottom) the targeting process.

that the 'windows' of chaotic transients seen in Fig. 3 are either approximately 200 or 2000 iterations in length without applying targeting, whilst with targeting they are either 10 or 12 iterations long.

6. CONCLUSIONS

We have described a new method for targeting which does not require knowledge of the governing equations and which can be adapted to cope with noise. A linearisation about an idealised trajectory is required which involves a lot of computation, but this can be done in advance so that very little computation is required during the actual targeting process, in contrast to other methods. A small parameter perturbation is then required at each iteration to counteract the effects of noise and to keep the orbit close to the idealised orbit.

The methods discussed here are for targeting in two-dimensional systems. Targeting is also possible in higher-dimensional systems (see e.g. [12,13]). The new targeting algorithms described here are easily generalised to the higher-dimensional problem, but more computation is involved in implementing the methods.

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